CNENGR115 Computers and Control

Transfer Function Models

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Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

Part 3 – Control Actions

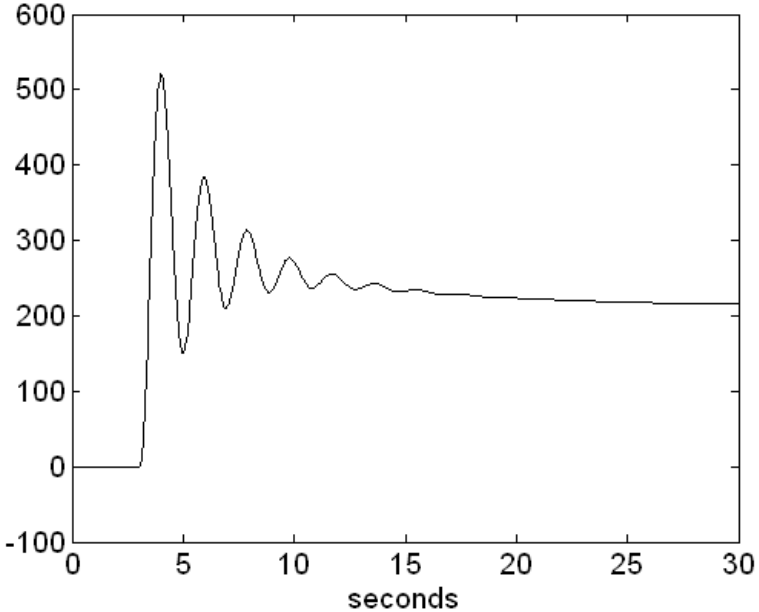
Generalised 1st Order Model

- We have developed a model for vehicle speed and converted it into a general **1st order** form
- Many other engineering systems can also be represented using this model

$$\tau\dot{x} + x = Ku(t)$$

Higher Order Systems

- Many engineering systems <u>cannot</u> be represented using a 1st order model
- In some cases, we require a **2nd order** model, e.g. the earlier equation for vehicle displacement (rather than speed) $m\ddot{y} + b\dot{y} = u(t)$
- Otherwise, 3rd or even higher order models are required e.g. torque of a **wind turbine**
- To handle such higher order systems we will use **Transfer Functions**



Notation for Rates of Change

- The rate of change of x is denoted by $\dot{x}$ or $\dfrac{dx}{dt}$
- The rate of change of $\dot{x}$ is denoted by $\ddot{x}$ or $\dfrac{d^2x}{dt^2}$
- The rate of change of $\ddot{x}$ is denoted by $\dddot{x}$ or $\dfrac{d^3x}{dt^3}$
- In general, we can use the following $\dfrac{d^nx}{dt^n}$

$$\left(n = 1, 2, 3, \ldots\right)$$

Notation for Differential Equations

- Consider again the **1st order** model, but with the new notation

$$\tau\dot{x} + x = Ku \quad \rightarrow \quad a_0\frac{dx}{dt} + a_1 x = b_0 u$$

- Similarly for a **2nd order** system

$$a_0\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_2 x = b_0 u$$

- Rates of change of the **input** signal are sometimes useful in the model, e.g.

$$a_0\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_2 x = b_0\frac{du}{dt} + b_1 u$$

Notation for Differential Equations

- **Generalised linear differential equation**

$$a_0 \frac{d^n x}{dt^n} + a_1 \frac{d^{n-1} x}{dt^{n-1}} + \ldots + a_{n-1} \frac{dx}{dt} + a_n x$$

$$= \quad b_0 \frac{d^m u}{dt^m} + b_1 \frac{d^{m-1} u}{dt^{m-1}} + \ldots + b_{m-1} \frac{du}{dt} + b_m u$$

- Examples:

$$a_0 \frac{dx}{dt} + a_1 x = b_0 u$$

$$a_0 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_2 x = b_0 u$$

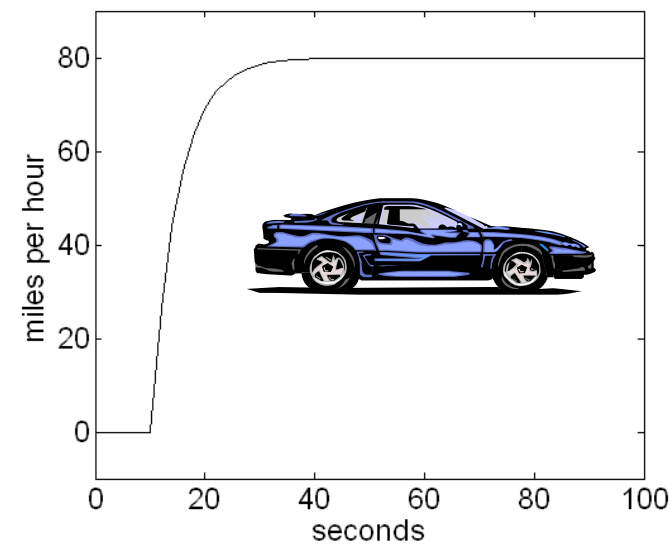
$$a_0 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_2 x = b_0 \frac{du}{dt} + b_1 u$$

Integration

- We can obtain an **analytical solution** to a differential equation by integration and hence find the time response
- Example – we have already used this formula for the **step response** of a first order system

$$x(t) = K\left(1 - e^{-t/\tau}\right)$$

- But what about solving for different inputs, e.g. sine wave, pulse and general time varying signals?
- And what about higher order systems?
- Engineers use **Laplace Transforms** and **Transfer Functions**
- These are also useful when simulation software such as **Matlab/Simulink** is used to find the numerical solution



Transfer Functions

- **Laplace Transforms** provide a tool for analytically solving differential equations
- **Transfer Functions** are based on Laplace Transforms with zero initial conditions
- Formally, we take Laplace Transforms and set the initial conditions to zero

$$x(t) \qquad X(s)$$

$$\frac{dx(t)}{dt} \text{ becomes } sX(s)$$

$$\frac{d^2x(t)}{dt^2} \text{ becomes } s^2X(s)$$

$$\frac{d^3x(t)}{dt^3} \text{ becomes } s^3X(s) \text{ and so on}$$

Transfer Functions

- In this introductory module, we will use a differential operator notation, as follows:

$$\frac{dx}{dt} \rightarrow sX$$

$$\frac{d^2x}{dt} \rightarrow s^2X$$

$$\frac{d^nx}{dt} \rightarrow s^nX$$

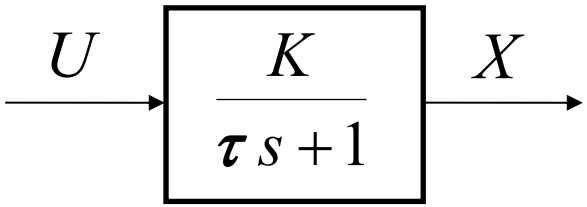
Example 1 – General 1st Order System

Taking the general 1st order model and using the notation above

$$\tau \frac{dx}{dt} + x = Ku \quad \rightarrow \quad \tau sX + X = KU \quad \rightarrow \quad (\tau s + 1)X = KU$$

$$X = \frac{K}{\tau s + 1} U$$

Transfer Function



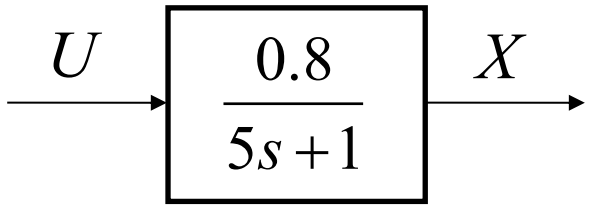
Example 2 – Vehicle Speed Model

Assuming $\tau = 5$ and $K = 0.8$ then the model for vehicle speed becomes:

$$5\frac{dx}{dt} + x = 0.8u \quad \rightarrow \quad 5sX + X = 0.8U \quad \rightarrow \quad (5s+1)X = 0.8U$$

$$X = \frac{0.8}{5s+1}U$$

Transfer Function



Example 3 – Higher Order System

$$a_0 \frac{d^3x}{dt^3} + a_1 \frac{d^2x}{dt^2} + a_2 \frac{dx}{dt} + a_3 x = b_0 \frac{d^2u}{dt^2} + b_1 \frac{du}{dt} + b_2 u$$

$$a_0 s^3 X + a_1 s^2 X + a_2 s X + a_3 X = b_0 s^2 U + b_1 s U + b_2 U$$

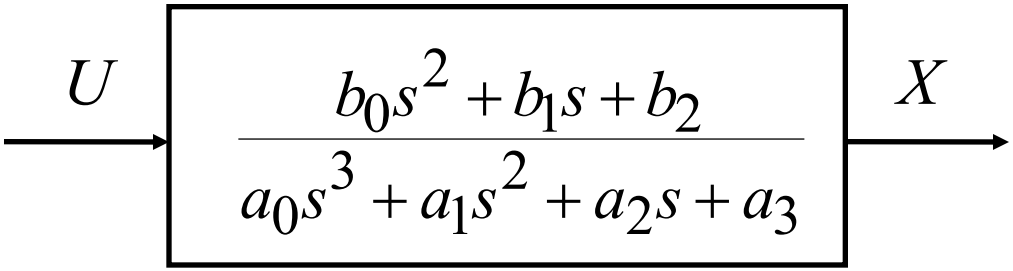
$$\left(a_0 s^3 + a_1 s^2 + a_2 s + a_3\right) X = \left(b_0 s^2 + b_1 s + b_2\right) U$$

$$X = \frac{b_0 s^2 + b_1 s + b_2}{a_0 s^3 + a_1 s^2 + a_2 s + a_3} U$$

Transfer Function

Example 3 – Higher Order System

$$X = \frac{b_0 s^2 + b_1 s + b_2}{a_0 s^3 + a_1 s^2 + a_2 s + a_3} U$$



Exercise

Rainfall
Nonlinear Element
Effective Rainfall
Dynamic Model
River Flow

© ROB HOWARD/PIN

Exercise

$$\mathrm{Y}(s) = \frac{0.01706s^2+0.1578s+0.001680}{s^2+0.3299s+0.001768}U(s)$$

- *Y*(*s*) is the **river flow** output
- *U*(*s*) is the "**effective rainfall**" input
- *s* is the differential operator, i.e. d/dt
- Obtained by statistical estimation from the hourly measured flow and effective rainfall

Determine the differential equation form of the model

Exercise

$$\mathrm{Y}(s) = \frac{0.01706s^2+0.1578s+0.001680}{s^2+0.3299s+0.001768}U(s)$$

$$(s^2 + 0.3299s + 0.001768)Y(s) = (0.01706s^2 + 0.1578s + 0.001680)U(s)$$

$$s^2Y(s) + 0.3299sY(s) + 0.001768Y(s) = 0.01706s^2U(s) + 0.1578sU(s) + 0.001680U(s)$$

Exercise

$$s^2Y(s) + 0.3299sY(s) + 0.001768Y(s) =$$
$$0.01706s^2U(s) + 0.1578sU(s) + 0.001680U(s)$$

$$\frac{d^2y(t)}{dt^2} + 0.3299\frac{dy(t)}{dt} + 0.001768y(t) =$$
$$0.01706\frac{d^2u(t)}{dt^2} + 0.1578\frac{du(t)}{dt} + 0.001680u(t)$$

Steady State Gain

- Recall that for first order systems, the **steady state gain** is given by ***K***

$$x(t \to \infty) = K \cdot u$$

- In general, the steady state gain is found by setting ***s* = 0** in the Transfer Function

Example 1 – General 1st Order System

$$\frac{K}{\tau s+1} \qquad \frac{K}{\tau \times 0+1} = K$$

Steady State Gain = K

$$\tau \dot{x} + x = Ku(t)$$

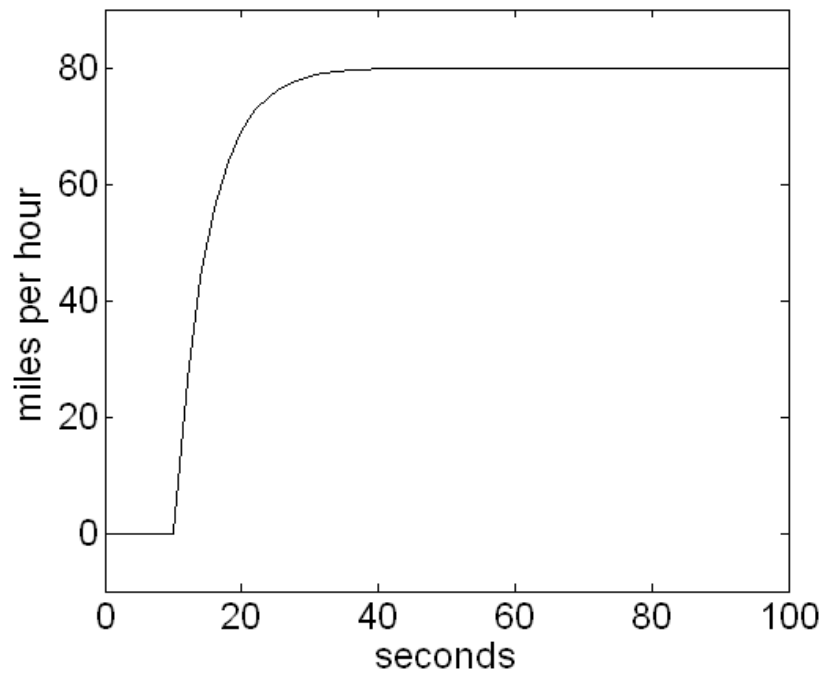
Example 2 – Vehicle Speed Model

$$X = \frac{0.8}{5s+1}U$$

Steady State Gain = 0.8

$$x(t \to \infty) = 0.8 \times u$$
$$= 0.8 \times 100 = 80 \text{ mph}$$

$$u = 100$$



$$5\dot{x} + x = 0.8u(t)$$

Example 3 – Higher Order System

$$X = \frac{b_0 s^2 + b_1 s + b_2}{a_0 s^3 + a_1 s^2 + a_2 s + a_3} U$$

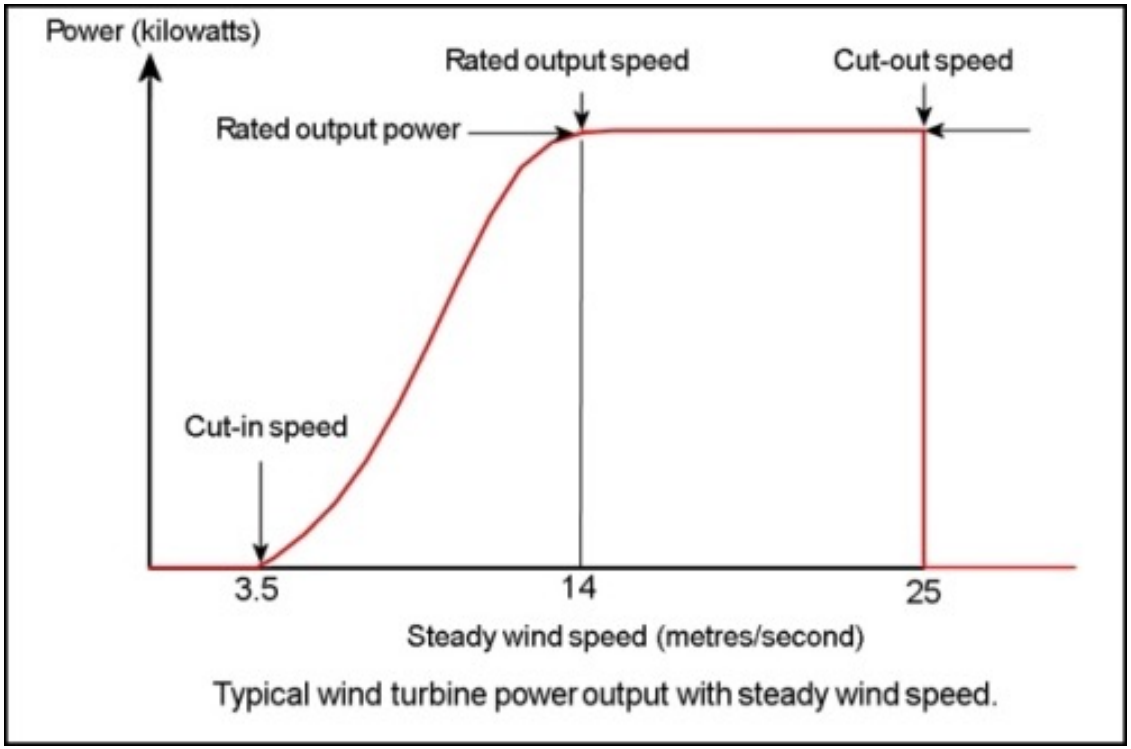
Steady State Gain = $\frac{b_2}{a_3}$

(value of the) output at steady state

$$x(t \to \infty) = \frac{b_2}{a_3} \times u$$

Case Study – Wind Turbine

- Aerodynamic Power Limiting
- Passive Control
- Active Control



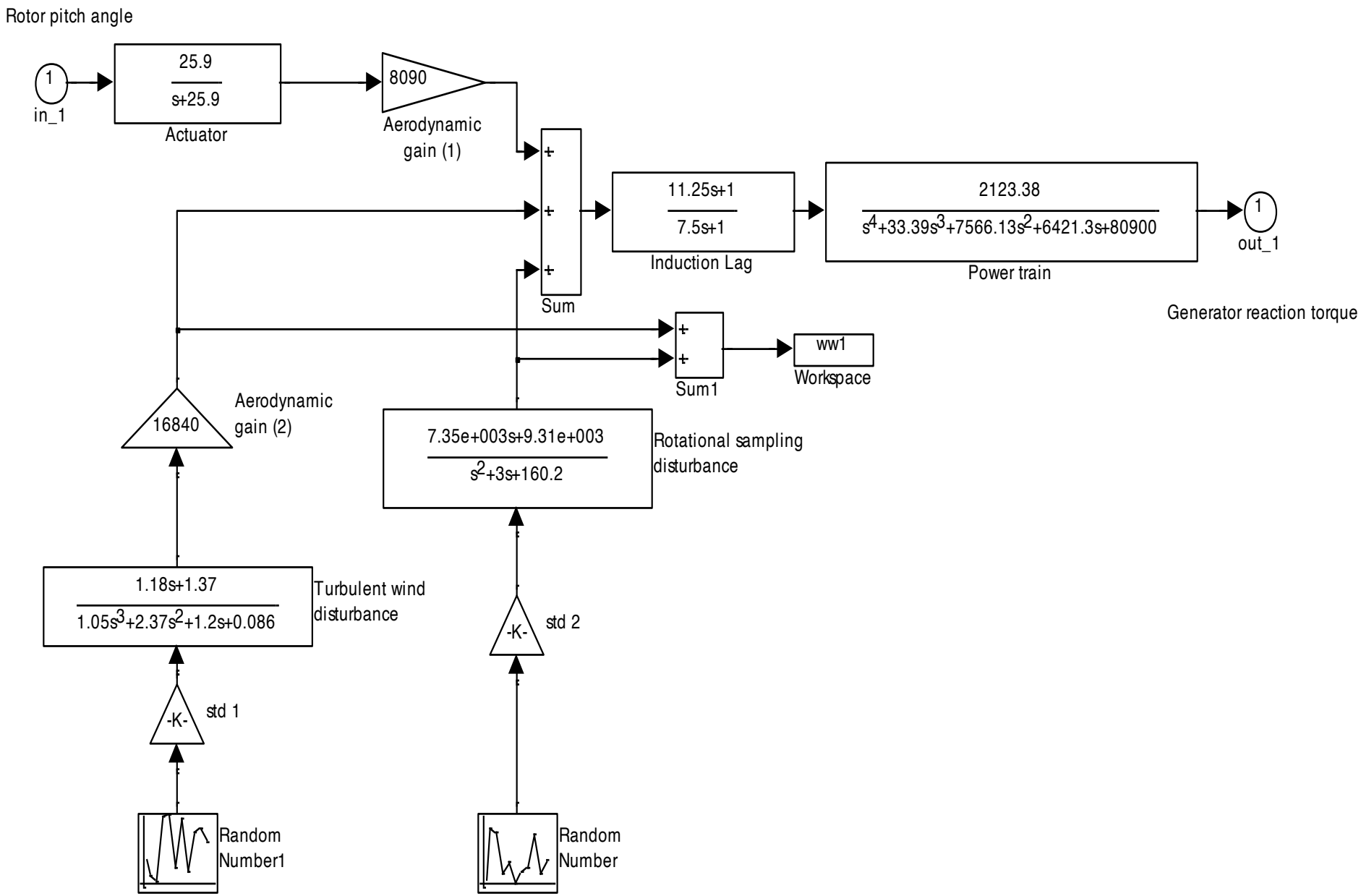
Typical wind turbine power output with steady wind speed.

Control Objectives

- **High level**
  - Reduce load transients throughout the wind turbine, so that the working life of the components are maximised
  - Smooth power output
  - Maximise energy capture
- **Low level**
  - Standard objectives (stability, satisfactory transient, etc.)
  - Maintain the demanded generator reaction torque by adjusting the blade pitch angle
  - Minimise disruption by wind disturbances

Wind Turbine Simulation

- Leithead, W.E., and Connor, B. (2000) Control of Variable Speed Wind Turbines: Dynamic Models, *International Journal of Control*, **73**, pp. 1173–1188.
- Includes discussion of a high order, multivariable and non–linear simulation model, surrogate for the real system
- On the next slide, a simpler, **linear** version of the model is constructed and solved using **Transfer Functions**

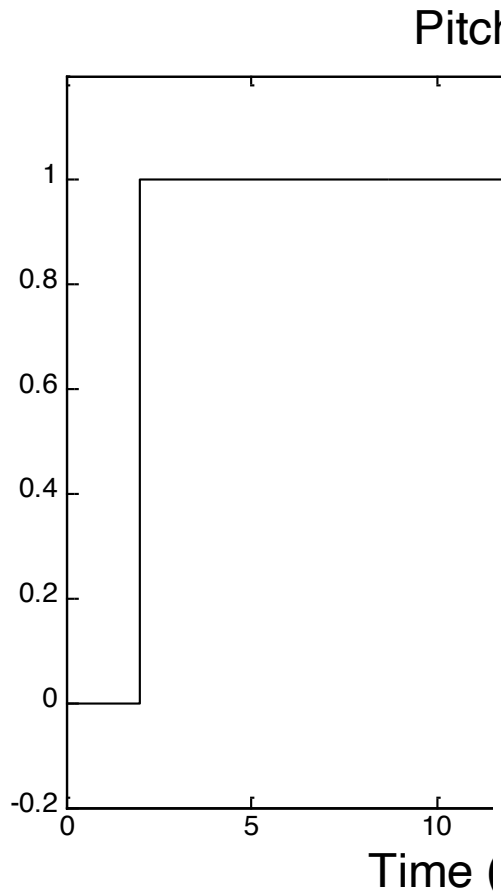

Rotor pitch angle
1
in_1
25.9
s+25.9
Actuator
8090
Aerodynamic gain (1)
Sum
11.25s+1
7.5s+1
Induction Lag
2123.38
s4+33.39s3+7566.13s2+6421.3s+80900
Power train
1
out_1
Generator reaction torque
Sum1
ww1
Workspace
16840
Aerodynamic gain (2)
7.35e+003s+9.31e+003
s2+3s+160.2
Rotational sampling disturbance
1.18s+1.37
1.05s3+2.37s2+1.2s+0.086
Turbulent wind disturbance
-K-
std 2
-K-
std 1
Random Number1
Random Number

Exercise

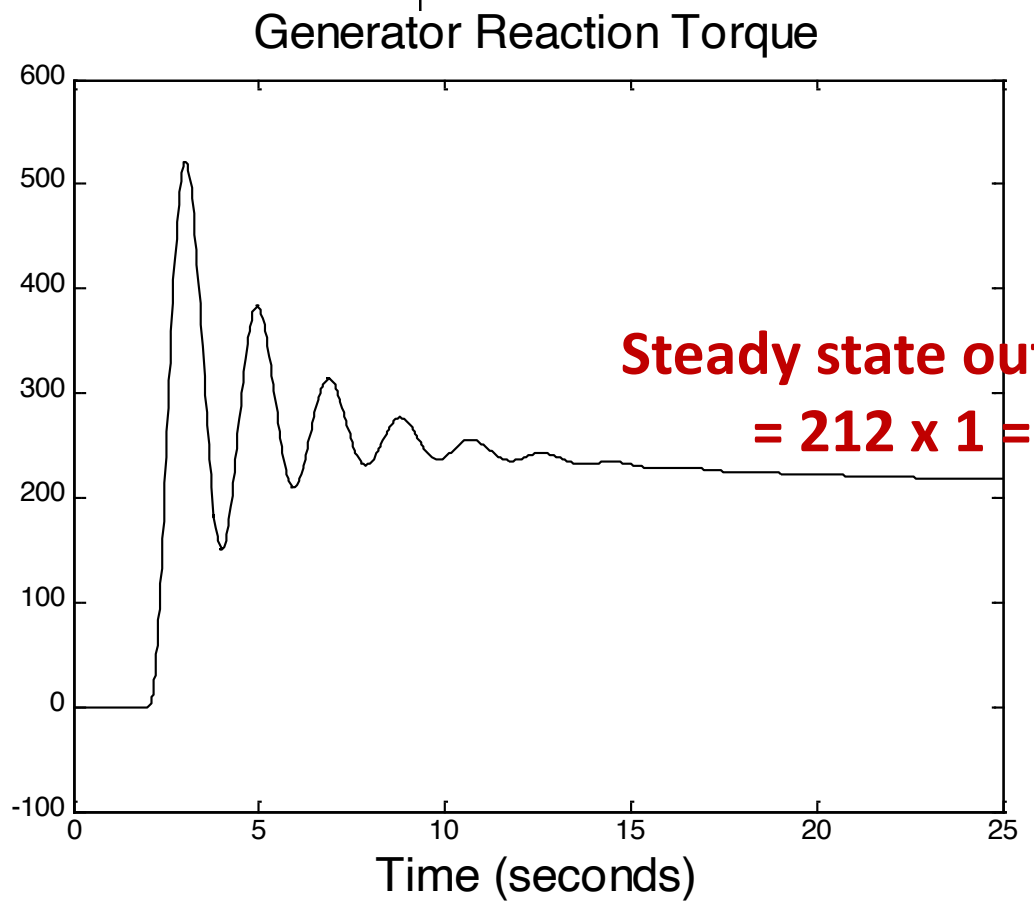
- Assuming zero wind disturbances (i.e. constant wind speed), what is the **steady state** relationship between the rotor pitch angle input and generator reaction torque output?
- And what is the **physical interpretation** of this result?
- Hints:
  - determine the steady state gain of the Actuator, Induction Lag and Power Train models (set **s = 0** in each case).
  - preview of one of the rules of block diagram analysis: **blocks in series are multiplied** (here you can multiply the steady state gains)
  - don't forget to include the Aerodynamic Gain (1) which is a scalar multiplier

Exercise

- Steady State Gain (SSG) of the **Actuator** Transfer Function is unity i.e. for s = 0, $\frac{25.9}{s+25.9} \rightarrow 1.0$
- SSG of the **Aerodynamic Gain** is 8090
- SSG of the **Induction Lag** is 1.0
- SSG of the **Power Train** is $\frac{2123.38}{80900} \rightarrow 0.026$
- These are all in series, hence the **overall SSG is 212**.
- **Physical interpretation**, steady state generator reaction torque = 212 x input



Input = 1.0



Steady state output = 212 x 1 = 212

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